MIDTERM EXAMINATION

Blockchain Technology, Smart Contracts and Digital Governance

Management Engineering – University of Palermo

Exam Format: Multiple-Choice Questions (MCQs) Total Questions: 40

SECTION 1 – BLOCKCHAIN FUNDAMENTALS

Understanding the structure, principles, and mechanics of blockchain technology.

Question 1. What is the primary characteristic that distinguishes blockchain from a traditional centralized database?

A. Blockchain distributes the maintenance and verification of records across multiple network participants.

B. Blockchain stores all information on one central server.

C. Blockchain eliminates the need for cryptographic security.

D. Blockchain allows only one authorized administrator to validate transactions.

Question 2. What is the fundamental purpose of cryptographic hashing in blockchain technology?

A. To produce a fixed-length digital fingerprint that helps detect alterations to data.

B. To encrypt every transaction so that no participant can read it.

C. To eliminate the need for consensus mechanisms.

D. To guarantee that all information entered into the blockchain is factually accurate.

Question 3. Which components are typically included in a blockchain block?

A. Transaction data, a reference to the previous block's hash, and other metadata.

B. Only the names and passwords of network participants.

C. Only the financial balance of the network administrator.

D. Only encrypted documents stored on a centralized server.

Question 4. What is the principal role of nodes in a blockchain network?

A. To participate in storing, validating, or communicating blockchain information according to their roles.

B. To replace the internet infrastructure used by organizations.

C. To guarantee that all transactions are legally enforceable.

D. To prevent users from accessing their transaction history.

Question 5. What is a consensus mechanism?

A. A protocol through which network participants agree on the valid state or ordering of blockchain transactions.

B. A legal contract signed by every blockchain user.

C. A centralized database management system.

D. A procedure that allows users to delete previously confirmed transactions.

Question 6. What is the main difference between Proof of Work (PoW) and Proof of Stake (PoS)?

A. PoW relies on computational work, while PoS selects validators based on staked assets and protocol rules.

B. PoW requires no energy, while PoS always requires industrial mining facilities.

C. PoW operates without cryptography, while PoS uses only traditional passwords.

D. PoW and PoS are identical mechanisms with different commercial names.

Question 7. Why is Proof of Stake generally considered more energy-efficient than Proof of Work?

A. It does not require validators to compete through energy-intensive computational mining.

B. It completely eliminates electricity consumption.

C. It operates exclusively on centralized servers.

D. It requires every transaction to be approved manually.

Question 8. What does immutability mean in the blockchain context?

A. Confirmed records are designed to be extremely difficult to alter without detection or violating network rules.

B. Every blockchain record can be freely modified by its creator.

C. All blockchain information automatically disappears after a fixed period.

D. Blockchain transactions can never be challenged in court.

Question 9. Why is blockchain frequently described as a technology that reduces reliance on trusted intermediaries?

A. Cryptography and consensus rules allow participants to verify records without depending entirely on one central authority.

B. Blockchain guarantees that every participant behaves ethically.

C. Blockchain removes all legal obligations between parties.

D. Blockchain prevents every possible form of fraud.

Question 10. What is a significant managerial disadvantage of implementing blockchain instead of a traditional database?

A. Greater complexity in governance, integration, performance, and operational costs may outweigh its benefits.

B. Blockchain cannot support any commercial transaction.

C. Blockchain necessarily prevents information sharing between organizations.

D. Blockchain always requires organizations to disclose their trade secrets publicly.

SECTION 2 – SMART CONTRACTS

Technical implications, legal aspects, and real-world applications.

Question 11. What is a smart contract in blockchain technology?

A. A program that automatically executes predefined operations when specified conditions are satisfied.

B. A traditional written agreement that must always be signed before a notary.

C. A verbal agreement between two cryptocurrency investors.

D. A digital document that cannot perform automated operations.

Question 12. What is the relationship between smart contracts and traditional legal contracts?

A. Smart contracts can automate contractual performance, but their legal validity and effects depend on applicable law and circumstances.

B. Every smart contract automatically replaces all legal agreements.

C. Smart contracts are legally binding in every jurisdiction regardless of their content.

D. Smart contracts eliminate the possibility of contractual disputes.

Question 13. Which is a realistic application of smart contracts outside cryptocurrency trading?

A. Automating insurance payments when reliable external data confirms an insured event.

B. Automatically guaranteeing the honesty of all insurance companies.

C. Replacing every human decision in the judicial system.

D. Eliminating the need to verify insurance claims or contractual conditions.

Question 14. What are gas fees on the Ethereum blockchain?

A. Fees paid for computational resources used to process transactions and execute smart contracts.

B. Government taxes imposed exclusively on cryptocurrency profits.

C. Fixed annual membership fees paid to Ethereum's central administrator.

D. Insurance premiums protecting users against all cyberattacks.

Question 15. Why can Ethereum gas fees fluctuate?

A. Because network demand and the computational resources required affect transaction costs.

B. Because gas fees are permanently fixed by international law.

C. Because gas fees depend exclusively on the nationality of users.

D. Because Ethereum transactions never compete for network resources.

Question 16. What is an oracle in a smart contract ecosystem?

A. A mechanism or service that supplies external information to blockchain applications.

B. A government agency that approves every blockchain transaction.

C. A cryptocurrency wallet used exclusively for anonymous payments.

D. A legal professional responsible for drafting all smart contracts.

Question 17. What is the so-called oracle problem?

A. Smart contracts may depend on external information whose accuracy and reliability cannot be guaranteed by blockchain consensus alone.

B. Blockchain cannot store any transaction information.

C. Smart contracts always require internet access from every individual user.

D. Blockchain validators are legally prohibited from using external data.

Question 18. How can smart contracts improve business processes?

A. By automating predefined operations, improving traceability, and potentially reducing administrative delays.

B. By eliminating all cybersecurity risks.

C. By guaranteeing that every business decision is profitable.

D. By removing the need for organizational governance.

Question 19. Why is human oversight still important when organizations use smart contracts?

A. Because coding errors, incorrect external data, legal disputes, and unforeseen circumstances may require human intervention.

B. Because smart contracts cannot execute any operation automatically.

C. Because blockchain systems never use predetermined rules.

D. Because human supervision makes cryptographic security unnecessary.

Question 20. Which situation best illustrates a legal risk associated with smart contracts?

A. An automatically executed transaction produces consequences inconsistent with the parties' legally enforceable agreement.

B. A company uses software to reduce repetitive administrative work.

C. A blockchain network verifies a valid transaction.

D. An organization improves transparency in its accounting processes.

SECTION 3 – BLOCKCHAIN APPLICATIONS AND CHALLENGES

Business innovation, supply chain management, scalability, and sustainability.

Question 21. How can blockchain improve supply chain management?

A. By providing a shared, traceable record of transactions and product movements among authorized participants.

B. By physically transporting products between warehouses.

C. By guaranteeing that all suppliers provide accurate information.

D. By eliminating the need for logistics managers.

Question 22. Which is a major challenge when implementing blockchain in supply chains?

A. Ensuring accurate data entry, interoperability, stakeholder cooperation, and effective governance.

B. Preventing companies from sharing any transaction information.

C. Eliminating all internet connections between suppliers.

D. Replacing every physical product with a digital token.

Question 23. What is the main purpose of using blockchain for food traceability?

A. To improve the ability to track product origins and movements across the supply chain.

B. To guarantee that food products can never be contaminated.

C. To eliminate every food safety inspection.

D. To replace agricultural production with automated software.

Question 24. What is Ethereum primarily known for in the blockchain ecosystem?

A. Supporting programmable smart contracts and decentralized applications.

B. Operating exclusively as a traditional centralized banking database.

C. Preventing developers from creating applications.

D. Requiring every transaction to be approved by a single government.

Question 25. What is a decentralized application (dApp)?

A. An application that uses smart contracts or decentralized network infrastructure for some of its operations.

B. An application that must always operate without an internet connection.

C. A program controlled exclusively by one national central bank.

D. A website that cannot communicate with blockchain networks.

Question 26. What does scalability mean in blockchain technology?

A. The ability of a blockchain system to accommodate increasing transaction demand while maintaining acceptable performance.

B. The physical size of the computers used by blockchain developers.

C. The number of legal agreements signed by cryptocurrency companies.

D. The ability to eliminate all transaction fees permanently.

Question 27. What is the main purpose of Layer 2 blockchain solutions?

A. To improve transaction processing efficiency by handling activity outside the base layer while relying on it for relevant security or settlement functions.

B. To replace blockchain with traditional paper records.

C. To remove all cryptographic verification.

D. To ensure that transactions can never be disputed.

Question 28. What does sharding aim to achieve in blockchain architecture?

A. To distribute data or processing responsibilities across separate parts of a system, potentially improving scalability.

B. To store all transactions permanently on a single personal computer.

C. To prohibit communication between blockchain participants.

D. To eliminate the need for network security.

Question 29. Which statement best describes blockchain's environmental impact?

A. Environmental impact depends on factors such as consensus design, energy sources, hardware, and network operation.

B. Every blockchain consumes exactly the same amount of energy.

C. Proof of Stake necessarily consumes more energy than Proof of Work.

D. Blockchain technology has no environmental impact under any circumstances.

Question 30. When should a company seriously consider adopting blockchain?

A. When multiple parties need a shared, verifiable record and the expected benefits justify governance, technical, and economic costs.

B. Whenever management wants to replace every existing information system.

C. Only when the company intends to issue its own cryptocurrency.

D. Whenever a centralized database would be simpler, cheaper, and fully sufficient.

SECTION 4 – GOVERNANCE, ETHICS AND FUTURE PERSPECTIVES

Data ownership, regulation, digital governance, and societal implications.

Question 31. How can blockchain contribute to digital identity management?

A. By supporting systems in which individuals can control identifiers and present verifiable credentials.

B. By guaranteeing complete anonymity in every digital interaction.

C. By eliminating the need for identity verification.

D. By automatically transferring ownership of personal data to governments.

Question 32. What is a potential privacy risk associated with public blockchains?

A. Permanently accessible transaction records may allow personal information or transaction patterns to be linked to identifiable individuals.

B. Public blockchains automatically delete all personal information.

C. Blockchain transactions cannot contain any personal data.

D. Cryptographic hashing always guarantees complete anonymity.

Question 33. Why can blockchain create challenges under the EU General Data Protection Regulation (GDPR)?

A. Because the persistence and distributed storage of personal data may complicate compliance with rights such as erasure and rectification.

B. Because the GDPR prohibits every use of blockchain technology.

C. Because blockchain records are automatically exempt from data protection law.

D. Because the GDPR applies only to paper-based information.

Question 34. Which principle is particularly important when designing blockchain systems that process personal data?

A. Data minimization, ensuring that personal data processing is limited to what is necessary for specified purposes.

B. Unlimited publication of all personal information.

C. Permanent retention of every document without justification.

D. Automatic disclosure of users' identities to all network participants.

Question 35. What is the principal objective of the EU Markets in Crypto-Assets Regulation (MiCA)?

A. To establish a harmonized regulatory framework for specified crypto-assets and related services within the European Union.

B. To prohibit every blockchain application in the European Union.

C. To replace the GDPR in all digital transactions.

D. To require all European businesses to accept cryptocurrency payments.

Question 36. What is the main ethical concern when blockchain governance is concentrated among a small number of participants?

A. Decision-making power may become concentrated, potentially undermining decentralization, accountability, and fair participation.

B. Concentrated governance automatically guarantees democratic representation.

C. Concentrated governance eliminates conflicts of interest.

D. Concentrated governance makes transparency unnecessary.

Question 37. Can blockchain technology succeed independently of cryptocurrencies?

A. Yes, because blockchain can support applications such as traceability, credential verification, and interorganizational recordkeeping.

B. No, because every blockchain must be used exclusively for cryptocurrency speculation.

C. No, because smart contracts cannot operate in business environments.

D. Yes, but only if blockchain systems abandon cryptographic verification.

Question 38. What is the potential role of blockchain in public administration?

A. Improving auditability, record integrity, and coordination in suitable public-sector processes.

B. Eliminating the need for administrative law.

C. Automatically guaranteeing that public officials never engage in misconduct.

D. Replacing every public institution with a private technology company.

Question 39. Which governance approach is most appropriate for a company adopting blockchain technology?

A. Combining technical security, legal compliance, risk management, clear responsibilities, and periodic auditing.

B. Allowing developers to operate without organizational oversight.

C. Ignoring privacy legislation because blockchain is decentralized.

D. Eliminating internal controls because transactions are recorded cryptographically.

Question 40. Which statement best describes the long-term transformative potential of blockchain?

A. Blockchain may improve coordination, verification, and trust across organizations when applied to appropriate use cases with effective governance.

B. Blockchain will necessarily replace every traditional database.

C. Blockchain guarantees the elimination of financial fraud and corruption.

D. Blockchain makes legal systems, professional judgment, and public institutions unnecessary.